SE4031 – Games Development

Assignment 1
2D Game Document

Game Title – quidditch2d

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Table of Contents

Overview 3
Implementation of Core Features 3
Screenshots 5
Control Guide 6
Creative Feature 6
Video Link 7

Overview

Quidditch2d is a 2D side-scrolling fantasy game inspired by magical themes. The player controls a young wizard flying through the sky on a broomstick. The main objective is to survive as long as possible while defeating incoming enemies using magical powers.The player continuously moves forward while enemies appear from different directions. The player must dodge obstacles and attack enemies using spells. The game becomes progressively more difficult as enemy speed and spawn rate increase over time.The player has a health system. Taking damage from enemies reduces health, and the game ends when health reaches zero. A score system tracks how long the player survives and how many enemies are defeated.

Implementation of Core Features

1. Flying System (Movement & Physics)

The player is always in a flying state instead of walking.

Upward Movement: Pressing a key (e.g., Space) moves the player upward smoothly.

Gravity Effect: When no input is given, the player slowly descends.

Smooth Control: Movement uses velocity instead of instant position changes for natural flying.

2. Spell Attack System

The player uses magical powers to defeat enemies.

Spell Casting: Clicking or pressing a key shoots a magical projectile.

Projectile Logic: Spells move forward and destroy enemies on collision.

Cooldown System: Small delay between spell usage to prevent spamming.

3. Enemy System

Enemies continuously spawn and attack the player.

Basic Enemies: Move toward the player in straight paths.

Damage System: Colliding with enemies reduces player health.

Difficulty Scaling: Enemy speed and spawn rate increase over time.

4. Game Manager (Core Logic Controller)

Handles overall game behavior.

Score Tracking: Increases over time and per enemy defeated.

Game Speed: Gradually increases difficulty.

Game Over Flow:

Stop gameplay

Show final score

Display restart/menu options

5. Health System

Player starts with fixed health.

Damage reduces health gradually.

Game ends when health reaches zero.

Screenshots

Control Guide

Keyboard/Spacebar – Thrust upward

Mouse/Left Click – Fire Gun

Creative Feature

AI Behavior: The Dark Seeker

The Dark Seeker acts as the main pressure mechanic in the game. Unlike normal enemies that move in simple paths, the Dark Seeker actively hunts the player, forcing continuous movement and strategic spell usage.

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1. Target Acquisition (Dynamic Tracking)

The Dark Seeker does not follow a fixed path. Instead, it constantly tracks the player's position.

- **The Logic:** Every frame, the enemy compares its position with the player's position and calculates the most direct path between them. As the player moves up or down while flying on the broomstick, the Dark Seeker instantly adjusts its direction to follow.

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2. Constant Pursuit Speed (Standardized Movement)

Homing enemies often move inconsistently depending on distance. The Dark Seeker avoids this issue.

- **The Logic:** A normalization technique is used to maintain a constant movement speed. Whether the enemy is far away or very close, it always moves at the same speed. This creates fair and predictable gameplay, allowing the player to react and dodge effectively.

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3. Free Movement (Omnidirectional Flight)

To match the magical flying theme, the Dark Seeker moves freely in all directions.

- **The Logic:** The enemy is not restricted to horizontal or vertical paths. It can move diagonally and adjust direction smoothly without needing rotation delays. This creates a fast and aggressive chasing behavior, increasing game difficulty.

Video Link

https://drive.google.com/drive/folders/1VdoJ2vuw9D6mzsI8m5dTxwiTy9v2cyrA?usp=sharing